Back To FORM @ TEMPLATE

1	Course Name:	Operating Systems				
	Course Code:	BCL1213				
	Course Classification:	Major (core)				
2	Synopsis:	This course provides an in-depth understanding of operating systems, including their design, functions, and user interfaces. The course emphasizes solving problems in distributed and concurrent systems using computing theory and mathematics. Students will also learn to implement software solutions within system constraints and predict system behavior under random events using probability and expectation.				
3	Name(s) of Academic Staff:	1	Ruffinus Rusell			
		2				
		3				
4	Semester and Year offered:	Year Offered	1	Semester	2	*Remarks:*
5	Credit Value:	3				
6	Pre-requisite/ co-requisite (if any):	NA				

7

Course Learning Outcomes (CLO)		
	CLO1	Discuss the fundamental concept of operating systems (C2, PLO1)
	CLO2	Apply knowledge of computing theory and mathematics to solve problems in distributed and concurrent systems. (C3, PLO2)
	CLO3	Demonstrate a simple software solution and test its performance across various operating systems (OS) using VirtualBox (P5, PLO3)

8 Mapping of the Course Learning Outcomes to the Programme Learning Outcomes, Teaching Methods and Assessment Methods

Course Learning Outcomes	Programme Learning Outcomes (PLO)												Teaching Methods	Assessment Methods
	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11			
CLO1	√												Discussion Forums, interactive Ebooks/recorded video lectures	eTest
CLO2		√											Virtual Simulations, Recorded Video Lectures with Hands-on Demos	Assignment
CLO3			√										Virtual Labs with VirtualBox, Interactive eBooks with Tasks	Practical assessment
Mapping with MQF Cluster of Learning Outcomes	C1	C2	C3A											
			C3D											

Indicate the primary causal link between the CLO and PLO by ticking '√' in the appropriate box.

C1 = Knowledge & Understanding, **C2** = Cognitive Skills, **C3A** = Practical Skills, **C3B** = Interpersonal Skills, **C3C** = Communication Skills, **C3D** = Digital Skills, **C3E** = Numeracy Skills, **C3F** = Leadership, Autonomy & Responsibility, **C4A** = Personal Skills, **C4B** = Entrepreneurial Skills, **C5** = Ethics & Professionalism

9 Transferable Skills (if applicable)

(Skills learned in the course of study which can be useful and utilized in other settings)

1	Cognitive skills
2	Digital Skills
3	

Open-ended response (if any)

4	

10 Distribution of Student Learning Time (SLT)

Note: This SLT calculation is designed for home grown programme only.

	Course Content Outline and Subtopics	CLO*	Learning and Teaching Activities**									Total SLT
			Face-to-Face (F2F)								NF2F Independent Learning (Asynchronous)	
			Physical				Online/ Technology-mediated (Synchronous)					
			L	T	P	O	L	T	P	O		
1	Introduction to Operating Systems -Overview of Operating Systems: Definitions, History, Types -Key Functions of Operating Systems: Process Management, Memory Management, File Systems -User Interfaces: Command Line Interfaces (CLI) and Graphical User Interfaces (GUI). - Lab session on exploring different operating systems (Windows, Linux, macOS)	1,2,3					2		4		6	
2	Process Management and Scheduling Processes: States, Control Blocks, Context Switching Process Scheduling: Algorithms (FCFS, SJF, RR, Priority Scheduling) Concurrency Issues: Deadlocks, Starvation	1,2					4	2			10	
3	Memory Management -Memory Hierarchy: Registers, Cache, Main Memory, Secondary Storage -Memory Allocation: Paging, Segmentation, Virtual Memory -Analysis of Memory Management Techniques in Modern OS	1,2					4	2			8	
4	Process Synchronization - Concurrency and the need for synchronization - Semaphores, mutexes, and condition variables - Deadlocks and deadlock prevention	1,2					3	2			6	
5	File Management -File System Structure: Directory Structures, File Allocation Methods -Storage Management: Disk Scheduling Algorithms, RAID, File System Implementation -Analysis of File Systems in Modern Operating Systems	1,2					3	2			6	
6	System Implementation and Constraints -System Constraints: CPU, Memory, Storage, Network -Implementing Software Solutions within System Constraints -Case Studies on Implementing Software in Resource-Constrained Environments	2,3					4		4		8	
7	Operating System Security and Protection -Security in Operating Systems: User Authentication, Access Control, Encryption -Protection Mechanisms: Policies, Domains, Access Control Lists -Analyzing Security Models in Modern Operating Systems	1,2,3					4		4		8	
8												
9												
10												
11												
12												
13												
14												
15												
16												
17												
18												
19												
20												
											SUB-TOTAL SLT:	**96**
			Face-to-Face (F2F)								NF2F	

	Continous Assessement	%	Physical	Online/ Technology-mediated (Synchronous)	Independent Learning for Assessment (Asynchronous)	
1	eTest	30		2	5.2	
2	Assignment	30		4	3.2	
3						
4						
5						
					SUB-TOTAL SLT:	**14.4**

	Final Assessement	%	Face-to-Face (F2F)		NF2F Independent Learning for Assessment (Asynchronous)	
			Physical	Online/ Technology-mediated (Synchronous)		
1	Practical Assessment	40		4	5.6	
2						
3						
4						
5						
					SUB-TOTAL SLT:	**9.6**

	SLT for Assessment:	24
	GRAND TOTAL SLT:	120
A	% SLT for F2F Physical Component: *[Total F2F Physical /(Total F2F Physical + Total F2F Online + Total Independent Learning) x 100)]*	
B	% SLT for Online & Independent Learning Component: *[(Total F2F Online + Total Independent Learning) /(Total F2F Physical + Total F2F Online + Total Independent Learning) x 100]*	**100.00**
C	% SLT for All Practical Component: *[% F2F Physical Practical + % F2F Online Practical]*	**10.00**
C1	% SLT for F2F Physical Practical Component *[Total F2F Physical Practical /(Total F2F Physical + Total F2F Online + Total Independent Learning) x 100)]*	
C2	*% SLT for F2F Online Practical Component* *[Total F2F Online Practical / (Total F2F Physical + Total F2F Online + Total Independent Learning) x 100]*	**10.00**

Please tick (√) if this course is **Industrial Training/ Clinical Placement/ Practicum** using 50% of Effective Learning Time (ELT)

Note:

* Indicate the CLO based on the CLO's numbering in Item 8

** For ODL programme: Courses with mandatory practical requiremnets imposed by the programme standards or any related standards can be exempted from complying to the minimum 80% ODL delivery rule in the SLT.

11	Identify special requirement or resources to deliver the course (e.g., software, nursery, computer lab, simulation room etc)	Virtualbox or any available tools for OS

12	References (include required and further readings, and should be the most current)	Khan, K., Operating Systems Fundamentals (2023). Kindle Edition, B0BSGB6GKT White, P., Operating System. (2024), Oryson Press, 9798887153292
13	Other additional information (if applicable)	

Note: Number of PLO indicated is purely for illustration purposes only and the number is subjected to the curriculum design.